MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, STRAIGHT-THRU, 30 THRU 110 CONTACT POSITIONS, CRIMP REMOVABLE CONTACTS (.100 SQ GRID)

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-55302.

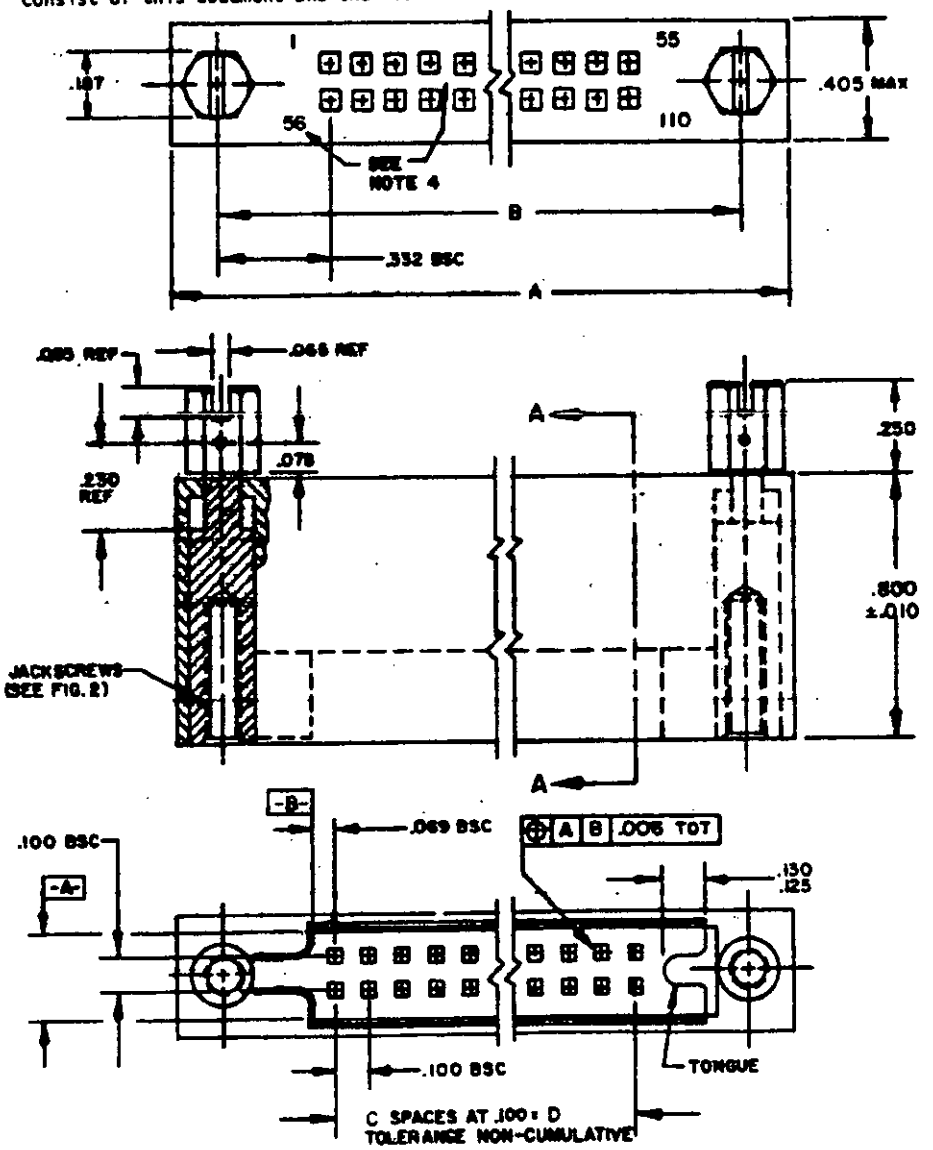
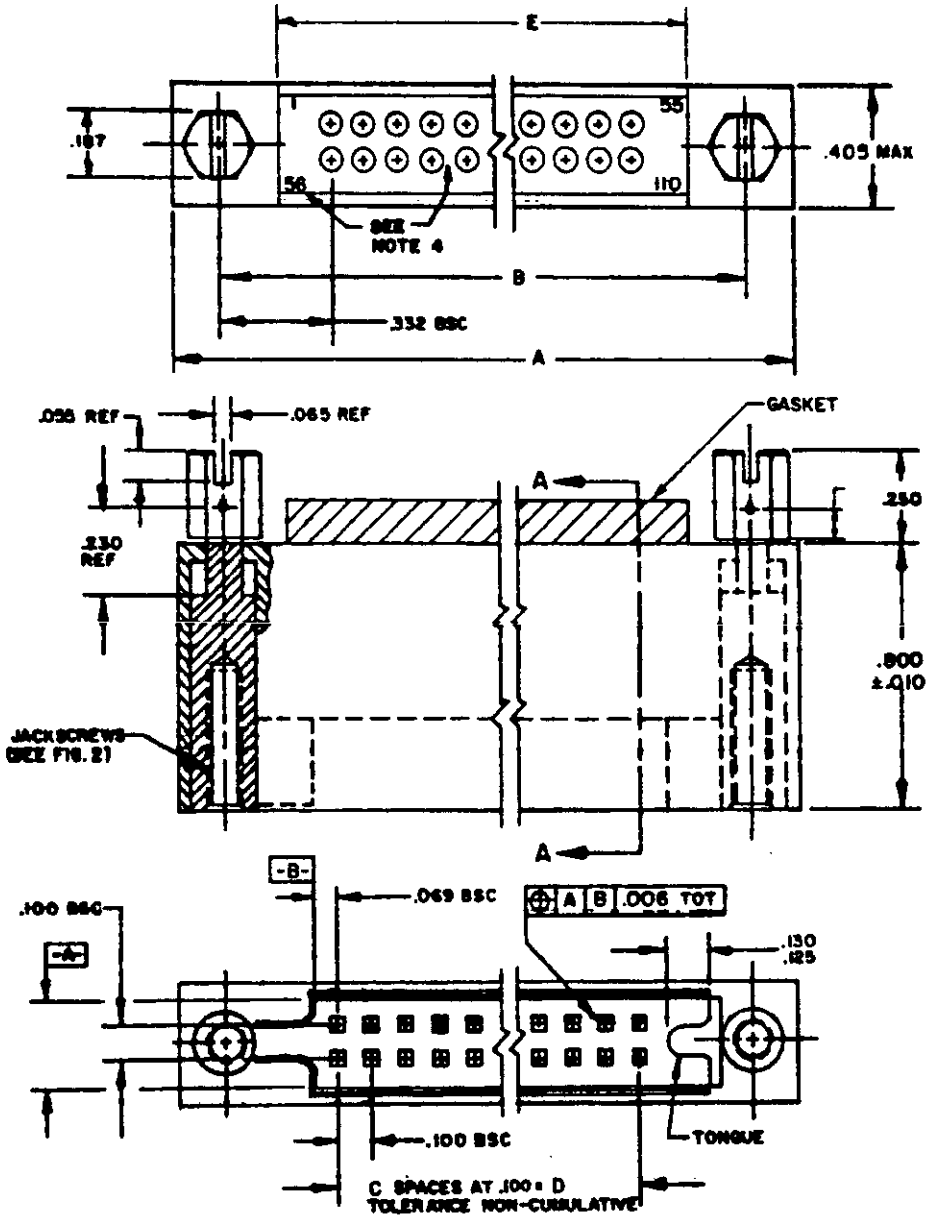


FIGURE 1. Connector, housing, plug (.100 square grid).

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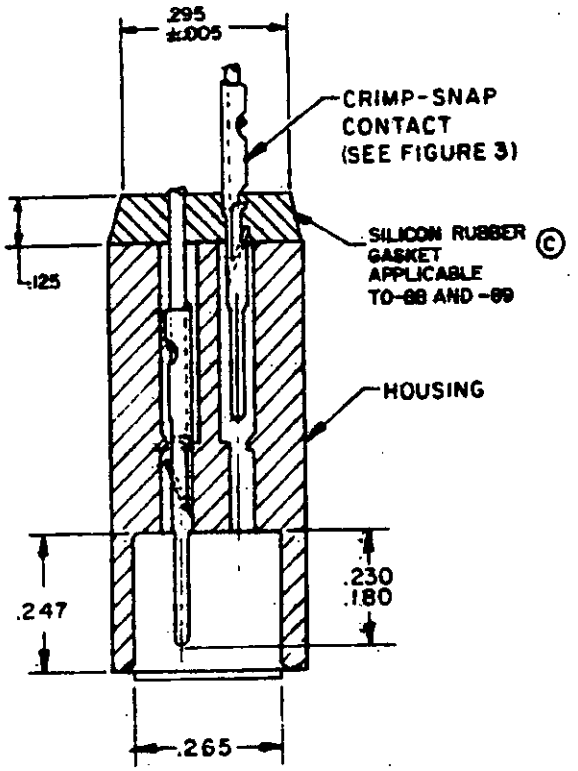
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© FIGURE 1. Connector, housing, plug, (.100 square grid) - Continued.

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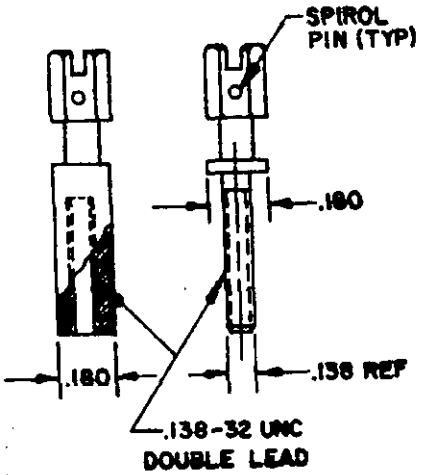
SECTION A-A

INCHES	MM
.005	.13
.006	.15
.010	.25
.015	.38
.055	1.40
.065	1.65
.069	1.75
.078	1.98
.100	2.54
.125	3.18
.130	3.30
.135	3.43
.140	3.56
.180	4.57
.187	4.75
.230	5.84
.247	6.27
.250	6.35
.265	6.73
.332	8.43
.400	10.16
.800	20.32
3.400	86.36
4.065	103.25
4.355	110.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals.
4. Numbers indicating groove end cavities and lines indicating every fifth cavity are stamped on housing.

FIGURE 1. Connector, housing, plug (.100 square grid) - Continued.

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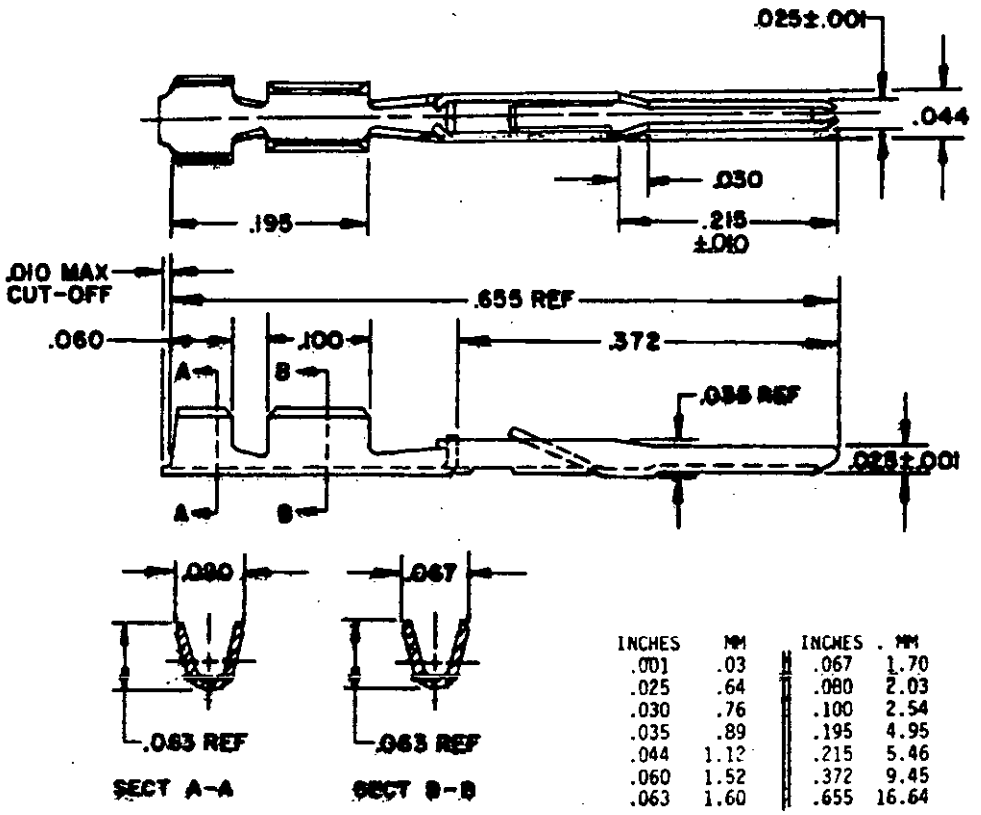


INCHES	MM
.138	3.51
.180	4.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals.

FIGURE 2. Turnable jackscrews.

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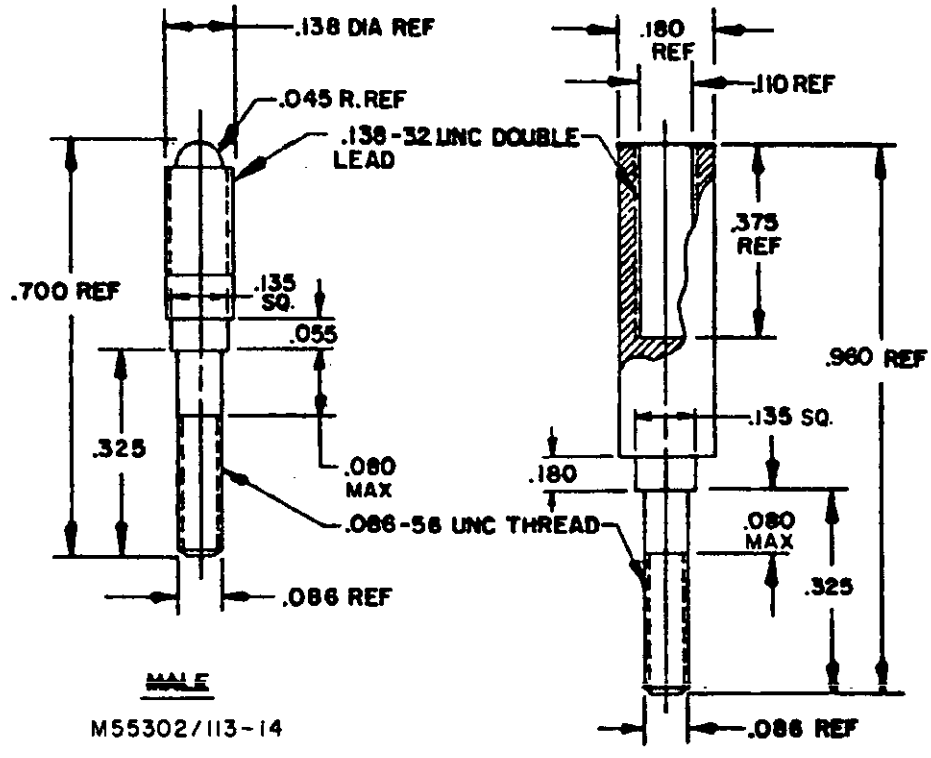


INCHES	MM	INCHES	MM
.001	.03	.067	1.70
.025	.64	.080	2.03
.030	.76	.100	2.54
.035	.89	.195	4.95
.044	1.12	.215	5.46
.060	1.52	.372	9.45
.063	1.60	.655	16.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals.
4. Contact crimp for use with (AWG) wire range no. 22 to no. 26, insulation range .036 (.91 mm) to .054 (1.37 mm).

FIGURE 3. Pin contact, crimp-snap.

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INCHES	MM	INCHES	MM
.045	1.14	.138	3.51
.055	1.40	.180	4.57
.080	2.03	.325	8.26
.086	2.18	.375	9.52
.110	2.79	.700	17.78
.135	3.43	.960	24.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals.
4. This connector mates with the external key type connectors specified in MIL-C-55302/27, and is equipped with mounting hardware, male and/or female (see table II).

FIGURE 4. Mounting hardware, for MIL-C-55302/27.

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© TABLE I. Housing dash numbers (without hardware), contact positions, and dimensions.

Dash number	Number of positions	Contact identification numbers	A	B	C	D	E
-15	30	1 15 16 30	2.355 (59.82)	2.065 (52.45)	14	1.400 (35.56)	N/A
-16	40	1 20 21 40	2.855 (72.52)	2.565 (65.15)	19	1.900 (48.26)	N/A
-17	50	1 25 26 50	3.355 (85.22)	3.065 (77.85)	24	2.400 (60.96)	N/A
-18	60	1 30 31 60	3.855 (97.92)	3.565 (90.55)	29	2.900 (73.66)	N/A
-19	70	1 35 36 70	4.355 (110.62)	4.065 (103.25)	34	3.400 (~~86.35~~)	N/A
-20	80	1 40 41 80	4.855 (123.32)	4.565 (115.95)	39	3.900 (99.06)	N/A
-21	90	1 45 46 90	5.355 (136.02)	5.065 (128.65)	44	4.400 (111.76)	N/A
-22	100	1 50 51 100	5.855 (148.72)	5.565 (141.35)	49	4.900 (124.46)	N/A
-23	110	1 55 56 110	6.355 (161.42)	6.065 (154.05)	54	5.400 (137.16)	N/A
-88	70	1 35 36 70	4.355 (110.62)	4.065 (103.25)	34	3.400 (86.36)	3.800 ±.010 (96.52) ±(.25)
-89	110	1 55 56 110	6.355 (161.42)	6.065 (154.05)	54	5.400 (137.16)	5.800 ±.010 (147.32) ±(.25)

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REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1 and table I.
Housing: Glass filled polyester per MIL-M-24519, type GPT-30F.
(C) Gasket - Silicon rubber.
Jackscrews and mounting hardware: Stainless steel in accordance with MIL-C-55302, passivated per ~~QQ-P-35~~.
Contacts: Brass, per QQ-B-626; gold plated per MIL-G-45204, type II, grade C, class 1; over nickel plating per QQ-N-290, class 2, 50 to 150 microinches.
Keying: See table II.
Crimp tool: AMP hand tool P/N 90346-1, or approved equivalent. For use with M55302/113-05 contacts.
AMP automatic tool P/N 466625-2, or approved equivalent. For use with M55302/113-06 contacts.
Contact identification: See figure 1.

Mating and unmating: The maximum mating force, in pounds, shall not exceed a value equal to 0.25 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of 0.025 times the number of contacts and shall not exceed the measured insertion force, when the housing is loaded with contacts and mated with connector specified in MIL-C-55302/27.

Current rating: 3.0 amperes maximum per contact.
2.25 amperes continuous per connector at room ambient.

Contact resistance: No individual contact pair shall have a resistance exceeding 0.020 ohms.

Contact retention: 3 pounds per contact.

Dielectric withstanding voltage:
Sea level: 900 volts rms.
High altitude: 200 volts rms.

(C) Part number: M55302/113-(dash number from tables I, II, and III.

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TABLE II. Dash number for pin header housing assembly. 1/

Assembly dash no.	Housing pin header	Keying jackscrews	
		Groove end	Tongue end
-7	-19	-11	-11
-8		-11	-12
-9		-12	-11
-10		-12	-12
-28	-15	-11	-11
-29		-11	-12
-30		-12	-11
-31		-12	-12
-36	-16	-11	-11
-37		-11	-12
-38		-12	-11
-39		-12	-12
-44	-17	-11	-11
-45		-11	-12
-46		-12	-11
-47		-12	-12
-52	-18	-11	-11
-53		-11	-12
-54		-12	-11
-55		-12	-12
-60	-20	-11	-11
-61		-11	-12
-62		-12	-11
-63		-12	-12
-68	-21	-11	-11
-69		-11	-12
-70		-12	-11
-71		-12	-12
-76	-22	-11	-11
-77		-11	-12
-78		-12	-11
-79		-12	-12
-84	-23	-11	-11
-85		-11	-12
-86		-12	-11
-87		-12	-12
-90	-88	-11	-11
-91		-11	-12
-92		-12	-11
-93		-12	-12
-94	-89	-11	-11
-95		-11	-12
-96		-12	-11
-97		-12	-12

1/ Pin header housing assembly consists of housing with jackscrews installed.

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TABLE III. Dash numbers for connector kits and component parts. 1/

Kit dash no.	Assembly pin header housing (see table II)	Hardware, receptacle mounting (see figure 4)			Contacts (loose piece) 2/
		-13	-14	2-56 nut	-05 3/
-01	-7	2	0	2	70
-02	-8	1	1		
-03	-9	1	1		
-04	-10	0	2		
-24	-28	2	0	2	30
-25	-29	1	1		
-26	-30	1	1		
-27	-31	0	2		
-32	-36	2	0	2	40
-33	-37	1	1		
-34	-38	1	1		
-35	-39	0	2		
-40	-44	2	0	2	50
-41	-45	1	1		
-42	-46	1	1		
-43	-47	0	2		
-48	-52	2	0	2	60
-49	-53	1	1		
-50	-54	1	1		
-51	-55	0	2		
-56	-60	2	0	2	80
-57	-61	1	1		
-58	-62	1	1		
-59	-63	0	2		
-64	-68	2	0	2	90
-65	-69	1	1		
-66	-70	1	1		
-67	-71	0	2		
-72	-76	2	0	2	100
-73	-77	1	1		
-74	-78	1	1		
-75	-79	0	2		
-80	-84	2	0	2	110
-81	-85	1	1		
-82	-86	1	1		
-83	-87	0	2		
-98	-90	2	0	2	70
-99	-91	1	1		
-100	-92	1	1		
-101	-93	0	2		
-102	-94	2	0	2	110
-103	-95	1	1		
-104	-96	1	1		
-105	-97	0	2		

1/ Component parts may be ordered separately.
2/ -06 contacts are supplied in reel form.
3/ Contacts shipped loose in a plastic bag, unless otherwise specified.

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